

Limiting Reactants and Percent Yield

Answer key with worked solutions

A. Understanding

1. Definitions.

(a) The reactant used up first, which limits the amount of product. (b) A reactant left over when the reaction stops. (c) The maximum amount of product, calculated from the limiting reactant.

2. Why yields are below 100%.

Any two of: the reaction does not go to completion; side reactions form other products; product is lost during separation or purification.

B. Working in moles

3. 3.0 mol H₂ + 2.0 mol O₂.

3.0 mol H₂ needs only 1.5 mol O₂, so (a) **H₂ is limiting**. (b) H₂O = **3.0 mol**. (c) O₂ left = 2.0 – 1.5 = **0.5 mol**.

4. 1.5 mol N₂ + 3.0 mol H₂.

1.5 mol N₂ would need 4.5 mol H₂, but only 3.0 mol is available, so (a) **H₂ is limiting**. (b) NH₃ = $3.0 \times \frac{2}{3}$ = **2.0 mol**. (c) N₂ used = 1.0 mol, so N₂ left = **0.5 mol**.

C. Working in grams

5. 5.40 g Al + 15.0 g Cl₂.

$n(\text{Al}) = 5.40 \div 26.98 = 0.2001 \text{ mol}$; $n(\text{Cl}_2) = 15.0 \div 70.90 = 0.2116 \text{ mol}$.

The Al would need $0.2001 \times \frac{3}{2} = 0.3002 \text{ mol Cl}_2$, but only 0.2116 mol is available, so (a) **Cl₂ is limiting**.

(b) $n(\text{AlCl}_3) = 0.2116 \times \frac{2}{3} = 0.1410 \text{ mol}$; $M(\text{AlCl}_3) = 133.33 \text{ g/mol}$; mass = **18.8 g**.

(c) Al used = 0.1410 mol; Al left = $0.2001 - 0.1410 = 0.0591 \text{ mol} \times 26.98 = \textbf{1.59 g}$.

6. Percent yield (15.2 g collected).

Percent yield = $15.2 \div 18.81 \times 100\% = 80.8\%$ (using the unrounded theoretical yield, 18.81 g).

7. 10.0 g CH₄ + 32.0 g O₂.

$n(\text{CH}_4) = 10.0 \div 16.04 = 0.6234 \text{ mol}$; $n(\text{O}_2) = 32.0 \div 32.00 = 1.000 \text{ mol}$. The methane needs 1.247 mol O₂, so (a) **O₂ is limiting**.

(b) $n(\text{CO}_2) = 1.000 \div 2 = 0.500 \text{ mol}$; mass = $0.500 \times 44.01 = 22.0 \text{ g}$.

(c) CH₄ used = 0.500 mol = 8.02 g; CH₄ left = $10.0 - 8.02 = 1.98 \text{ g}$.

D. Challenge

8. A 108% yield.

Most likely the product was still **wet** or contained **impurities**, which added mass. Mass cannot be created, so a pure, dry product can never exceed the theoretical yield.